

VDA11.6

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Exercise 6 (Conservative Vector Fields, 7 Points)

By definition, a vector field $\mathbf{v}$ is conservative if it can be written as the gradient of some continuously differentiable scalar field ϕ , $\mathbf{v} = \nabla\phi$.

a) Argue why any conservative vector field $\mathbf{v}$ is divergence-free, $\nabla \cdot \mathbf{v} = 0$, or provide a counterexample. (2P)

This statement is false. For example, $\phi(x, y, z) = x^2 + y^2 + z^2$ has $\mathbf{v} = \nabla\phi = \begin{pmatrix} 2x \\ 2y \\ 2z \end{pmatrix}$, and $\nabla \cdot \mathbf{v} = \Delta\phi = 2 + 2 + 2 = 6 \neq 0$.

b) Argue why any three-dimensional conservative vector field $\mathbf{v}$ is irrotational, $\nabla \times \mathbf{v} = \mathbf{0}$, or provide a counter example. (2P)

If $\nabla \times \mathbf{v}$ exists, we would have that $\nabla \times \mathbf{v} = \nabla \times \nabla\phi = \begin{vmatrix} i & j & k \\ \frac{d}{dx} & \frac{d}{dy} & \frac{d}{dz} \\ \frac{d\phi}{dx} & \frac{d\phi}{dy} & \frac{d\phi}{dz} \end{vmatrix} = \begin{pmatrix} \frac{d^2\phi}{dydz} - \frac{d^2\phi}{dzdy} \\ \frac{d^2\phi}{dzdx} - \frac{d^2\phi}{dxdz} \\ \frac{d^2\phi}{dxdy} - \frac{d^2\phi}{dydx} \end{pmatrix}.$

As ϕ is continuously differentiable, Schwarz's theorem guarantees that if the second partial derivatives of ϕ exist, they are symmetric. Thus $\frac{d^2\phi}{dxdy} = \frac{d^2\phi}{dydx}$, etc. We deduce that $\nabla \times \mathbf{v} = \mathbf{0}$.

c) Which of the following two vector fields is conservative? How could you tell? Write down a corresponding scalar potential ϕ . (3P)

We can compute the curls of the given vector fields:

$$\nabla \times \begin{pmatrix} y^2 + z \\ x^2 - z \\ x^2 - y \end{pmatrix} = \begin{vmatrix} i & j & k \\ \frac{d}{dx} & \frac{d}{dy} & \frac{d}{dz} \\ y^2 + z & x^2 - z & x^2 - y \end{vmatrix} = \begin{pmatrix} 0 \\ 1 - 2x \\ 2x - 2y \end{pmatrix}$$

As the curl of the first vector field is not identically zero, it cannot be a conservative vector field. Regarding the second:

$$\nabla \times \begin{pmatrix} y + z \\ x - z \\ x - y \end{pmatrix} = \begin{vmatrix} i & j & k \\ \frac{d}{dx} & \frac{d}{dy} & \frac{d}{dz} \\ y + z & x - z & x - y \end{vmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

So the curl of this field is identically zero and could be a conservative vector field. We will search a potential function whose gradient is $\mathbf{v}_2$. Such a function would verify $\phi(x, y, z) = \phi(0, 0, 0) + \int_0^x \frac{d\phi}{dx}(s, 0, 0)ds + \int_0^y \frac{d\phi}{dy}(x, s, 0)ds + \int_0^z \frac{d\phi}{dz}(x, y, s)ds$. We can assign $\phi(0, 0, 0) = 0$, and the partial derivatives of ϕ would coincide with the components of its gradient $\mathbf{v}_2$. Thus:

$$\begin{aligned} \phi(x, y, z) &= \int_0^x \frac{d\phi}{dx}(s, 0, 0)ds + \int_0^y \frac{d\phi}{dy}(x, s, 0)ds + \int_0^z \frac{d\phi}{dz}(x, y, s)ds = \\ &= \int_0^x 0 ds + \int_0^y x ds + \int_0^z x - y ds = 0 + yx + zx - zy = xy + xz - yz \end{aligned}$$

And indeed it is easy to check that $\nabla\phi = \begin{pmatrix} y + z \\ x - z \\ x - y \end{pmatrix} = \mathbf{v}_2(x, y, z)$, so $\mathbf{v}_2$ is indeed a conservative vector field.